Finding the Band Gap and Observing the Hall Effect

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A configured system of a circuit board, power supply, thermistor and multi-meters were used to demonstrate the band gap and hall effects of several germanium crystals. The band gap energy of germanium crystal number 586851 was experimentally found by extraction of a linear fit of the calculated conductivity, which resulted in $E_g = (0.50 \pm 0.04)\ eV$. The hall effect was investigated for two given crystals by observing the behavior of the extracted density in order to find the carrier type, with crystal number 586353 found to be an n-type semi-conductor with a carrier density of $\rho = (8.43 \pm 0.84)10E20\ m^{-3}$, and crystal number 586352 was found to be a p-type semi-conductor with a carrier density of $\rho = (-7.75 \pm 0.78)10E20\ m^{-3}$.

I. INTRODUCTION

Semi-conductors have proved useful since their discovery, with germanium and silicon being common elements known to have the properties of a semi-conductor which can be altered.[4] Demonstration of the band theory can reveal that the electron states in a crystal act as bands of energy, separated by gaps. A small impurity, also known as "doping", can allow an electrical current to be sent through a crystal in order to uncover the energy of it's band gap. Additionally, the carrier type and density in semi-conductor crystals can be mathematically determined once a magnetic field is applied. The carrier type depends on what kind of impurity the crystal possesses. A semi-conductor is noted as p-type if it contains holes, while it will be noted as n-type if it contains electrons as charge carriers. [5]

II. INVESTIGATING THE BAND GAP

Firstly, crystal number 586851 was placed in the circuit board apparatus. The equipment was connected in accordance to the diagram1 from the lab manual[3]. Once the configuration of the apparatus was confirmed to be correct, the HP6234A power supply will be turned on and set to 4 V for the full duration of the experiment. In order to expose the temperature dependence[2] of the band gap, the current will be increased on the heater. Since increasing the current does not instantaneously increase temperature, voltage measurements will need additional time to resolve fluctuations from the increasing heat, 8 minutes will be allotted between increased heater current values.

While taking voltage measurements for each current value, only the first, third, and fifth current measurements will be repeated in order to

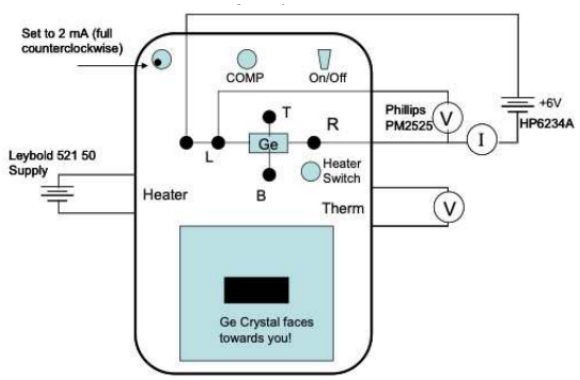


FIG. 1. Schematic diagram of the apparatus of the experiment in the configuration necessary to investigate the band gap energy of crystal number 585681.

get an uncertainty. The standard deviation of these measurements in each trial will be taken, and the maximum uncertainty found between all trials will be applied as a single voltage uncertainty. With recorded values of voltage and current, the resistance can be calculated by ohms law

$$R = V/I \tag{1}$$

where the constant voltage is from the power supply. The resistance has a propagated uncertainty equation of

$$\sigma_R = \sqrt{(\frac{\sigma_v}{I})^2 + (\frac{\sigma_I V}{I^2})^2}. \tag{2}$$

The resistance has a relation to conductivity which is

$$\sigma_0 = L/AR, \tag{3}$$

with L being the length of the crystal and A being the cross sectional area. The temperature can be calculated from the relation

$$T(K) = 273 + 100 * V \tag{4}$$

with V as voltage and the resulting temperature in Kelvin. Once the temperature is acquired, the conductivity of the crystal can be calculated with

$$\sigma = \sigma_0 * e^{\frac{-E_g}{2k_b T}} \tag{5}$$

with the Boltzmann constant in joules per kelvin. This equation can be log-transformed to extract the energy as a slope of a linear fit by

$$ln(\sigma) = -E_g * \frac{1}{2K_b T} + ln(\sigma_0). \tag{6}$$

To conduct a linear fit and extract the band gap energy, a least squares fitting will be used. The reciprocal of temperature will be plotted against the log conductivity. This will reveal a slope in the linear form

$$y = mx + b, \tag{7}$$

which then will finally extract the band gap energy

$$E_g = -2mK_b. \tag{8}$$

A. Findings of the Band Gap

The band gap energy of the germanium crystal found is $E_g = (0.42 \pm 0.03)\ eV$. This is much different than the accepted value of $E_g = 0.67\ eV$.[1] The reduced chi-squared value from the fit is $\chi_r^2 = 0.90$ which is near the desired value of 1 and indicates this is a good fit for the data, however it is possible the errors in the results slightly over estimate the actual observed deviation. It is noticeable on the resulting graph3

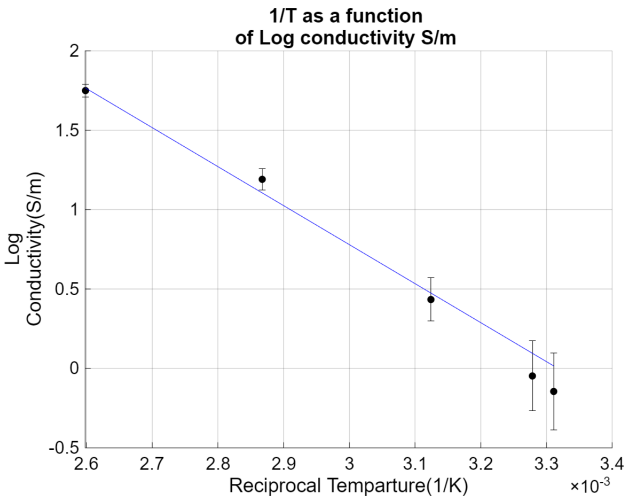


FIG. 2. Reciprocal temperature as a function of log conductivity with error bars and a least squares fitting. The fit equation is $ln(\sigma) = (-2.46 \pm 0.17)10E03 * (1/T) + (8.15 \pm 0.46)$. The band gap energy is $E_g = (0.42 \pm 0.03)\ eV$.

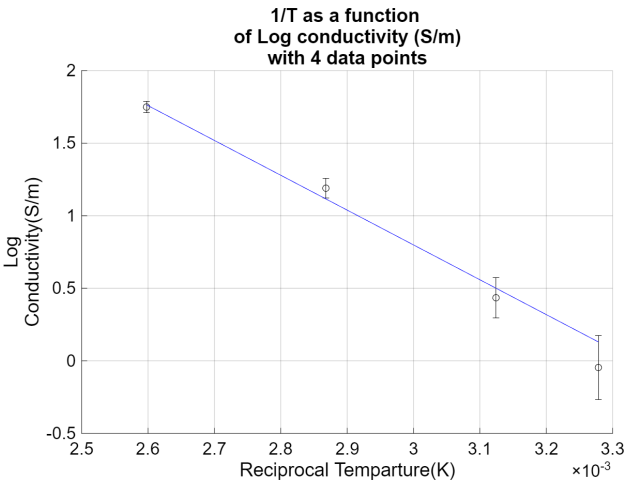


FIG. 3. Reciprocal temperature as a function of log conductivity of only 4 data points with error bars and a least squares fitting. The fit equation is$ln(\sigma) = (-2.40 \pm 0.19)10E03 * (1/T) + (8.00 \pm 0.50)$. The band gap energy is$E_g = (0.41 \pm 0.03)\ eV$.

that the last two data points are very close together. Another fit will be produced with only the first four data points in hopes of eliminating any errors with the slope calculation.

Attempting another calculation did not resolve any errors. The bang gap energy calculated with omitting the last data point is $E_g = (0.42 \pm 0.03)\ eV$, which is the same as the first calculation. The reduced chi-squared value from this fit is $\chi_r^2 = 1.09$ which is also near 1. This indicates the data is likely a good fit, however there are still errors present. Due to the persistent errors causing erroneous results, all measurements will be redone with changes in the method of collecting data. The first measurements were taken in 8 minute increments allowing the heater to settle, however the new method will have only 2 minute increments. The new data sets will be applied the same way as the previous one to find the band gap energy.

The band gap energy using the remeasured data set is $E_g = (0.50 \pm 0.04)\ eV$. This is closer to the accepted value than the last results, alas it still does not agree. The reduced chi-squared value of the new fit is $\chi_r^2 = 0.02$, which is very low. This can indicate the fit is too good to be true, despite using the same method of calculation which may be caused by not enough data points, or the distribution of points is coincidentally too good of a fit. The errors causing values far from the accepted values could come from a number of sources. Only 5 data points were taken to ensure the heater current is spread over a wide range of values, however it may have been

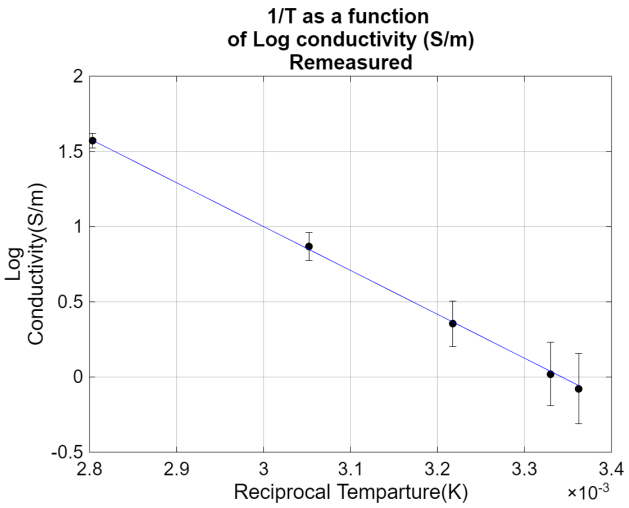


FIG. 4. Second data set reciprocal temperature as a function of log conductivity with error bars and a least squares fitting. The fit equation is $ln(\sigma) = (-2.92 \pm 0.23)10E03 * (1/T) + (9.75 \pm 0.68)$. The band gap energy is $E_g = (0.50 \pm 0.04)\ eV$.

better to take a larger quantity of data points. It is also possible the equipment had precision or calibration flaws that may have added additional errors into the data.

III. INVESTIGATING THE HALL EFFECT

The equipment will be re-configured in accordance to the diagram in the lab manual5 to conduct an investigation of the hall effect. Two crystals with numbers 586852 and 586853 will each be investigated to find the carrier type and density. A magnetic field sensor will also be calibrated and placed between the coil field and crystal to collect data. Measurements of magnetic field will be taken by varying the current

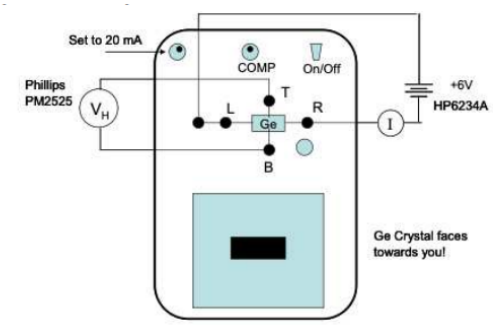


FIG. 5. Schematic diagram of the apparatus of the experiment in the configuration necessary to investigate the hall effect.

on the magnetic power supply. Uncertainty for the collected variables will be done by repeated measurements and the standard deviation formula. In order to find the carrier type of each crystal, the equation

$$p = -\frac{IB}{qV_h t} \tag{9}$$

reveals the number of positively charged carriers with t as the thickness of the crystal, and B as the measured magnetic field. If V_h is positive, the charge carriers are electrons which indicates an n-type crystal. Otherwise, if the value of V_h is negative, the charge carriers are positive and indicative of a p-type crystal. This distinction is due to the Lorentz force depending on both the charge and the velocity of the carriers in the crystal.[5] When doing a linear fit, the number density of the carriers can be extracted with slope m as

$$n = \frac{I}{qmt}. \tag{10}$$

A. Findings of the Hall Effect

For the first crystal measured, number 586853, the carrier density is $\rho = (8.43 \pm 0.84)10E20\ m^{-3}$, and the slope of the linear fit is a positive value, $(14.7 \pm 0.04)10E-02\ V/T$ indicating an n-type carrier. However, the reduced chi-Squared value for this fit is $\chi_r^2 = 0.24$, which is low compared to the desired value of 1. This could indicate an overestimation of errors in the data. The second crystal, number 586853 has a carrier density of $\rho = (-7.75 \pm 0.78)10E20\ m^{-3}$, and a slope value of $(-16.02 \pm 0.02)10E-02\ V/T$, which indicates the crystal is a p-type carrier. The reduced chi-squared value for the fit of the second crystal is $\chi_r^2 = 25$, which is very large and could indicate the fit is not good for the data, or the errors were underestimated. The densities and carrier types of both crystals were able to be determined despite errors in the χ_r^2 values despite the figures67 visually showing good linear fits.

Multiple systemic errors could have occurred during the measurement process such as equipment limitations or insufficient uncertainty collection. Despite the undesired values of reduced chi-squared for each fit, the carrier types of each crystal have been identified.

IV. CONCLUSION

A given germanium crystal was firstly investigated to find the band gap energy. The first calculated value was $E_g = (0.42 \pm 0.03)\ eV$, which was far from the accepted value of $E_g = 0.67\ eV$. To attempt to resolve errors, another calculation was done differently by omitting the last data point that was thought to be a source of error, which caused the same resulting band gap energy. For both calculation methods the χ_r^2 value was near 1. After revising calculations and acquiring new measurements in order to troubleshoot potential errors, the band gap energy was found to be $E_g = (0.50 + / - 0.04)\ eV$, now closer, but 4 sigma away. The reduced chi-squared value for the new data was notably low. Between the first and remeasured data set, the amount of time allocated to the heater settling at a temperature was modified, which allowed the

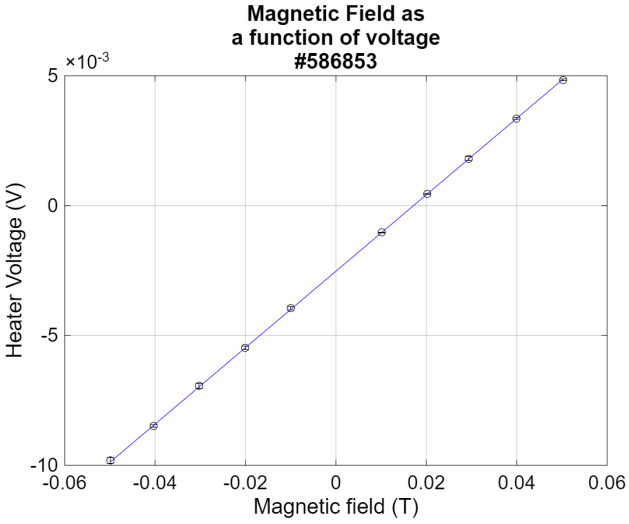


FIG. 6. Magnetic field as a function of voltage through crystal 586853 with error bars and a least squares fitting. The fit equation is $V = (14.71 \pm 0.05)10E-02 * T + (250 \pm 0.08)10E-04$. The carrier density is $\rho = (8.43 \pm 0.84)10E20\ m^{-3}$.

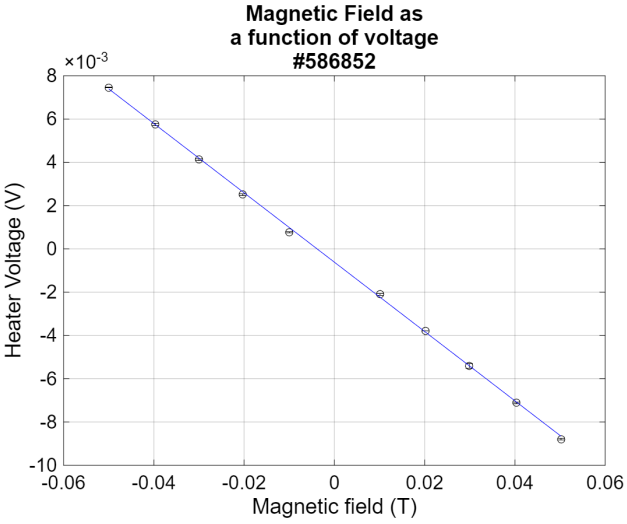


FIG. 7. Magnetic field as a function of voltage through crystal 586852 with error bars and a least squares fitting. The fit equation is $V = (-16.02 \pm 0.02)10E-02 * T + (-6.13 \pm 0.07)10E-04$. The carrier density is $\rho = (-7.75 \pm 0.78)10E20\ m^{-3}$.

result to become closer to the accepted value. This change did not completely absolve all result errors. It is likely that additional systemic errors contributed to the inaccurate value.

When investigating the hall effect for two new crystals, it was found that crystal number 586353 is a n-type semi-conductor with a carrier density of $\rho = (8.43 \pm 0.84)10E20\ m^{-3}$, and crystal 586352 is a p-type semi-conductor with a carrier density of $\rho = (-7.75 \pm 0.78)10E20\ m^{-3}$. However, the resulting linear fits of the data resulted in poor χ_r^2 values despite visually linear results. The poor reduced chi-squared values could indicate estimation errors of the uncertainties, or the fits are not sound in accordance to the data. Sources of error may have been from equipment limitations or calibration errors.

[1] Practical Electron Microscopy and Database An Online Book, Second Edition by Dr. Yougui Liao (2006) `https://www.globalsino.com/EM/page2063.html`

[2] Lautenschlager, P., Allen, P. B., and Cardona, M. (1985). Temperature dependence of band gaps in Si and Ge. Physical Review B, 31(4), 2163.

[3] University of Minnesota Physics 3605W Lab Manual, Band Gap and Hall effect, (2024)

[4] Jenkins, T. (2005). A brief history of... semiconductors. Physics education, 40(5), 430-439.

[5] Chien, C. (Ed.). (2013). The Hall effect and its applications. Springer Science and Business Media.